Genetic drift, with and without mutation

Exercise sheet Journal club

Model. A population carries N copies of a gene, each of type A or type a, and x is the fraction of type A. Neither type is fitter. Each generation,

1. every copy mutates to the other type, independently, with probability u;
2. the next generation is N copies drawn at random, with replacement, from the mutated pool (Wright–Fisher resampling).

Time t is counted in generations. For large N, with Nu held fixed, $x(t)$ is close to a diffusion on $0 < x < 1$. Throughout,

$$\theta = 2Nu.$$

Beware a clash of names: biologists call the randomness of resampling *genetic drift*. It is the noise B, not the drift A.

Convention: as in the talk, B is the variance per unit time, $\langle(\Delta x)^2\rangle = B\,\Delta t$. Parts (a)–(d) are the core exercise; part (e) is an optional extension. Equation numbers refer to Gardiner, 2nd edition.

(a) Build the forward equation

Given x, find the mean and the mean square of the change Δx over one generation. Keep the leading order in $1/N$, counting u as order $1/N$. Show that

$$A(x) = u\,(1 - 2x), \qquad B(x) = \frac{x(1-x)}{N},$$

and write the forward equation for $p(x,t)$ and the backward operator $\mathcal{L}^\dagger$.

- Which term comes from mutation, and which from resampling?
- Why does B vanish at $x = 0$ and $x = 1$, and why does A not?
- Roughly how many generations does it take for x to change by an amount of order one?

Hint: After mutation the fraction is $x' = x(1-u) + (1-x)u$. Resampling makes N times the new fraction a binomial variable, with N trials and success probability x'.

(b) Find the stationary state

1. Use the potential solution (5.2.39) to show that

$$p_s(x) = \frac{\Gamma(2\theta)}{\Gamma(\theta)^2}\,[x(1-x)]^{\theta-1}.$$

   Sketch it for $\theta = \frac{1}{4}$, 1 and 4. What competes with what? Why is p_s U-shaped when mutation is rare?

2. Classify the boundary $x = 0$ with Feller's test (5.2.25)–(5.2.28). Show that it is regular for $0 < \theta < 1$ and an entrance boundary for $\theta \geq 1$.

3. When $x = 0$ is regular, a boundary condition has to be chosen. Which one does the discrete model choose, and why? Why does that make the potential solution the right one?

4. What does Gardiner's quick rule (5.2.20)–(5.2.24) predict at $x = 0$? Why does it fail? Compare the squared-radius example from the talk, $\mathrm{d}y = (dD - 2ky)\,\mathrm{d}t + \sqrt{4Dy}\,\mathrm{d}W$: near $x = 0$, what plays the role of the dimension d?

Hints: Near $x = 0$, keep only the leading power of x in A, in B, and in Feller's functions f and g. The normalisation is a Beta integral, $\int_0^1 [x(1-x)]^{\theta-1}\,\mathrm{d}x = \Gamma(\theta)^2/\Gamma(2\theta)$ (§§5.2.1, 5.2.2).

(c) Find the eigenfunctions

The backward operator is $\mathcal{L}^\dagger f = u(1-2x)\,f' + \dfrac{x(1-x)}{2N}\,f''$.

1. Show that $\mathcal{L}^\dagger$ maps a polynomial of degree n to a polynomial of degree n. Deduce that each eigenfunction Q_n is a polynomial of degree n, and read off from the coefficient of x^n that
$$\lambda_n = \frac{n(n-1+2\theta)}{2N}, \qquad n = 0, 1, 2, \ldots$$
2. Find Q_1 and Q_2.
3. Show that $\langle x(t)\rangle - \frac{1}{2} = (x_0 - \frac{1}{2})\,e^{-2ut}$ exactly. Why does N not appear? Which rate is the spectral gap?
4. Here $P_n = p_s\,Q_n$. Why? The heterozygosity $H = 2x(1-x)$ is the probability that two copies drawn at random differ. Find its stationary average,
$$H_s = \frac{\theta}{1+2\theta},$$
without doing an integral.
5. Interpret the two terms of $\lambda_2 = 1/N + 4u$.

Hints: Try $Q_2 = x(1-x) + c$. The relation $P_\lambda = p_s Q_\lambda$ comes from (5.2.71)–(5.2.76). Since $Q_0 = 1$, biorthogonality says $\int P_n\,\mathrm{d}x = 0$ for $n \geq 1$ (5.2.83).

(d) Switch off mutation

Now set $u = 0$, so $A = 0$ and $B = x(1-x)/N$. Label the modes by their degree, as in (c), so here the absorption rate is λ_2 and the rate of approach to the QSD is $\lambda_3 - \lambda_2$ (in the talk, λ_1 and $\lambda_2 - \lambda_1$).

1. Show that there is no stationary density. What type are the ends now? What happens to the population?
2. Now $\lambda_n = n(n-1)/2N$, so $\lambda_0 = \lambda_1 = 0$. Using $Q_1 = x$, show that $\langle x(t) \rangle = x_0$ for all t. Deduce the probability that type A takes over the whole population, and check it against the splitting probability (5.2.186)–(5.2.190).
3. $Q_2 = x(1-x)$ decays at $\lambda_2 = 1/N$. Show that $P = Q/[x(1-x)]$ solves the forward equation whenever Q solves the backward one. Compare $P_\lambda = p_s Q_\lambda$: what plays the role of p_s? Deduce that *the QSD is uniform on* $(0,1)$. Normalise so that $\int P_2 Q_2 \, \mathrm{d}x = 1$, and show that the probability of keeping both types is $S(t) \simeq 6x_0(1-x_0)\, e^{-t/N}$ at long times.
4. Unlike the $\sin(\pi x/L)$ QSD in the talk, this one does not vanish at the absorbing ends. Find the current out through each end, and check that the total is λ_2. Why can p stay finite at an end here?
5. Check that $Q_3 = x(1-x)(1-2x)$, and find λ_3. At what rate does p/S approach the QSD? What changes if $x_0 = \frac{1}{2}$?
6. Find the mean time $T(x_0)$ until one type is lost, from (5.2.154), and evaluate $T(\frac{1}{2})$. Explain why the average of T over the QSD must be $1/\lambda_2 = N$. Check it.

Hints: With $A = 0$, (5.2.154) reads $\frac{1}{2}BT'' = -1$. You may need $\int_0^1 x \ln x \, \mathrm{d}x = -\frac{1}{4}$. What happens to $p(x,t)$ if it starts as the QSD? (§§5.2.5, 5.2.7, 5.2.8)

(e) Optional: from one to the other

Let mutation be rare, $\theta \to 0$ at fixed N.

1. Show that $p_s \to \frac{1}{2}\delta(x) + \frac{1}{2}\delta(x-1)$, in the sense that a fixed neighbourhood of each end holds probability $\frac{1}{2}$. Show that $\lambda_1 \to 0$. Which conserved quantity of (d) does $Q_1 = 1 - 2x$ become?

2. Show that
$$\frac{p_s\, Q_2}{\theta/2} \;\longrightarrow\; 1 - \tfrac{1}{2}\delta(x) - \tfrac{1}{2}\delta(x-1).$$
Its interior is the QSD of (d). What do the δ terms mean? Check that this limit is orthogonal to both zero modes of (d), $Q_0 = 1$ and $Q_1 = x$.

3. When $\theta \ll 1$, the population almost always consists of a single type. New mutants appear at rate Nu per generation, and by (d) each takes over with probability $1/N$. Show that this two-state picture relaxes at exactly λ_1. So neutral variants replace each other at the mutation rate, whatever the population size: Kimura's result, the basis of the molecular clock. Compare the double well in the talk. Which two time scales must be well separated, and when does the picture fail?

Hints: As $\theta \to 0$, $\Gamma(2\theta)/\Gamma(\theta)^2 \simeq \theta/2$, and $\int_0^\varepsilon x^{\theta-1}\,\mathrm{d}x = \varepsilon^\theta/\theta$. Hold ε fixed while $\theta \to 0$.

The Wright–Fisher diffusion is additional material beyond the book examples. References: M. Kimura, *Solution of a process of random genetic drift with a continuous model*, Proc. Natl. Acad. Sci. USA **41**, 144 (1955), which solves part (d); M. Kimura, *Diffusion models in population genetics*, J. Appl. Probab. **1**, 177 (1964); M. Kimura, *Evolutionary rate at the molecular level*, Nature **217**, 624 (1968), for part (e). The accompanying key uses the same part labels.